

MUSAFEN TABLET 10MG

DESCRIPTION: A round 7 mm diameter yellow tablet with markings "DUOMA/DUOMA" and "d".

COMPOSITION: Each tablet contains Baclofen 10 mg.

PHARMACODYNAMICS:

The precise mechanism of action of baclofen has not been fully determined. It acts mainly at the spinal cord level to inhibit the transmission of both monosynaptic and polysynaptic reflexes, possibly by hyperpolarization of primary afferent fiber terminals resulting in antagonism of the release of putative excitatory transmitters (i.e. glutamic and aspartic acids). Actions at supraspinal sites may also be involved.

Baclofen is an effective antispastic agent with a spinal site of action. Its mechanism of action and pharmacological properties are different from those of other antispastic agents.

Baclofen depresses monosynaptic and polysynaptic reflex transmission, probably by various actions, including stimulation of GABAB-receptors. This stimulation in turn inhibits the release of excitatory amino acids (glutamate and aspartate) in guinea pig preparations. Neuromuscular transmission is not affected by baclofen.

Baclofen exerts an antinociceptive effect. The clinical significance of this awaits clarification. In neurological diseases associated with spasm of the skeletal muscles, the clinical effects of Baclofen take the form of a beneficial action on reflex muscle contractions and of marked relief from painful spasm, automatism, and clonus. Baclofen, where indicated, improves the patient's mobility, making for greater independence, and facilitating passive and active physiotherapy. Baclofen stimulates gastric acid secretion.

PHARMACOKINETICS:

Baclofen is rapidly and completely absorbed from the gastrointestinal tract. Maximum concentrations of unchanged drug are attained in plasma in 2 to 4 hours after an oral dose.

The distribution volume of baclofen amounts to 0.7 litre/kg. In cerebrospinal fluid the active substance attains concentrations approx. 8.5 times lower than in plasma.

The plasma elimination half-life of baclofen averages 3 to 4 hours. It is bound to plasma proteins to the extent of about 30%. Baclofen is eliminated largely in unchanged form. Within 72 hours, approx. 75% of the dose is excreted via the kidneys, approx. 5% of this quantity being in the form of metabolites. The remainder of the dose, including 5% as metabolites is excreted in the faeces. The main metabolite is beta-chlorophenyl-gamma-hydroxybutyric acid, which is pharmacologically inactive.

INDICATION:

Spasticity of the skeletal muscles in multiple sclerosis. Spastic conditions occurring in spinal cord diseases of infectious, degenerative, traumatic, neoplastic or unknown origin, eg. spastic spinal paralysis, amyotrophic lateral sclerosis, syringomyelia, transverse myelitis, traumatic paraplegia or paraparesis and compression of the spinal cord; muscle spasm of cerebral origin, especially where due to infantile cerebral palsy. In spasticity due to cerebrovascular accident or degenerative or neoplastic brain disease, Musafen tablet is less suitable owing to the likelihood of intolerance, but it may be tried if administered cautiously. Not recommended in Parkinsons Disease or spasticity arising from strokes.

RECOMMENDED DOSAGE:

Dosage regimen

Treatment should be initiated with small doses of Musafen Tablet, then gradually increased. It is recommended that the lowest effective dose be used. The optimum dosage must be individually adapted to the patient's requirements so that clonus, flexor and extensor spasms, and spasticity are reduced, while avoiding adverse effects as far as possible.

In order to prevent excessive weakness and falling, Musafen Tablet must be used with caution when spasticity is needed to maintain upright posture and balance in locomotion or other functions. It may be important to maintain a certain degree of muscle tone and allow occasional spasms to help support circulatory function and, possibly, prevent deep vein thrombosis.

Close monitoring is required at the start of treatment so that potential adverse effects such as general muscle weakness, abrupt loss of muscle tone (risk of falling), fatigue or confusion can be rapidly detected, and the dosage adjusted as necessary.

If baclofen is to be withdrawn after prolonged administration (longer than 2-3 months), the dose should be reduced gradually over the course of about 3 weeks except in overdose-related emergencies, or where serious adverse effects have occurred (see section WARNINGS & PRECAUTIONS), since abrupt withdrawal may occasionally be associated with anxiety and confusional states, hallucinations, cerebral seizures, dyskinesia, tachycardia or hyperthermia.

Withdrawal should also be gradual on account of the risk of a transient rebound effect involving increased spasm frequency and/or severity (see section WARNINGS & PRECAUTIONS).

Treatment with Musafen Tablet should be re-evaluated if it has not proved successful within 6 to 8 weeks of the patient's reaching the maximum recommended dose.

Adults

Treatment should be started with a dosage of 15mg daily, given in 3 divided doses, then gradually increased. Depending on the response, each of the divided doses should subsequently be increased by 5 mg every three days. 30-80 mg per day is considered the standard daily dose, but 100 to 120 mg may be given in isolated cases (patients under medical supervision in hospitals).

Special dosage instructions

Children and adolescents

Experience with Musafen Tablets is limited in children and adolescents. The minimum age for treatment is 12 months.

Treatment usually starts with very low amounts, e.g., 0.3 mg/kg/day, given in 4 divided doses. The dosage should be cautiously raised, at intervals of about 1-2 weeks, until it becomes sufficient for the individual patient's requirements. The daily dosage for maintenance therapy is 0.75-2 mg/kg. However, in patients over 10 years of age, a maximum daily dose of 2.5 mg/kg may be given.

The dose should not exceed 40 mg/day in children below 8 years of age, but a maximum dose of 60 mg/day may be given in children over 8 years of age.

Elderly patients (aged 65 years or above) and patients with spasticity of cerebral origin

The dosage should be increased particularly slowly in elderly and weakened patients suffering from organic brain disorders, cardiovascular disease, respiratory failure, or hepatic or renal impairment, and also in patients with spasticity of cerebral origin.

Renal impairment

Musafen Tablet should be used with caution in patients with renal impairment. As baclofen plasma concentrations are elevated in patients undergoing chronic haemodialysis, a particularly low dose of Musafen Tablet (approx. 5 mg/day) should be selected. Symptoms of overdose have been reported with doses of over 5 mg per day in this clinical situation (see SYMPTOMS AND TREATMENT OF OVERDOSE).

Musafen Tablet should not be used in patients with end stage renal failure unless the expected benefit of further treatment outweighs the potential risk. These patients must be closely monitored for early signs of overdose (e.g. drowsiness, lethargy; see sections WARNINGS & PRECAUTIONS and SYMPTOMS AND TREATMENT OF OVERDOSE).

ROUTE OF ADMINISTRATION: For oral use. Musafen Tablet should be taken at mealtimes with a small amount of liquid.

CONTRAINDICATIONS:

Known hypersensitivity to baclofen or to any of the excipient.

WARNINGS & PRECAUTIONS:

Psychiatric and nervous system disorders

Patients suffering from psychotic disorders, schizophrenia, depressive or manic disorders, confusional states or Parkinson's disease, should be treated cautiously with Musafen Tablet and kept under careful surveillance, because these conditions may become exacerbated.

Suicide and suicide-related events have been reported in patients treated with baclofen. In most cases, the patients had additional risk factors associated with an increased risk of suicide including alcohol use disorder, depression and/or a history of previous suicide attempts. Close supervision of patients with additional risk factors for suicide should accompany

therapy with Musafen Tablet. Patients (and caregivers of patients) should be alerted about the need to monitor for clinical worsening, suicidal behavior or thoughts or unusual changes in behavior and to seek medical advice immediately if these symptoms present.

Epilepsy

Special attention should be given to patients known to suffer from epilepsy since lowering of the convulsion threshold may occur and seizures have occasionally been reported in connection with the discontinuation of Musafen Tablet or with overdosage. Adequate anticonvulsive therapy should be continued and the patient should be carefully monitored.

Others

Musafen Tablet should be used with caution in patients with, or with a history of, peptic ulcers, as well as in patients with cerebrovascular diseases or with respiratory or hepatic impairment.

Since adverse effects are more likely to occur, a cautious dosage schedule should be adopted in elderly and patients with spasticity of cerebral origin (see section RECOMMENDED DOSAGE).

Renal impairment

Musafen Tablet should be used with caution in patients with renal impairment and should be administered to end-stage renal failure patients only if the expected benefit outweighs the potential risk (see section RECOMMENDED DOSAGE).

Neurological signs and symptoms of overdose including clinical manifestations of toxic encephalopathy (e.g., confusion, somnolence, hallucination) have been observed in patients with renal impairment taking Musafen Tablet at doses of more than 5mg per day. Patients with renal impairment should be closely monitored for prompt diagnosis of early signs and symptoms of toxicity (see section SYMPTOMS AND TREATMENT OF OVERDOSE).

Particular caution is required when combining Musafen Tablet with drugs or medicinal products which may significantly impact renal function. Renal function should be closely monitored and Musafen Tablet daily dosage adjusted accordingly to prevent baclofen toxicity.

Besides discontinuing treatment, unscheduled hemodialysis might be considered as a treatment alternative in patients with severe baclofen toxicity. Hemodialysis effectively removes baclofen from the body, alleviates clinical symptoms of overdose and shortens the recovery time in these patients.

Urinary disorders

On Musafen Tablet treatment, neurogenic disturbances affecting the emptying of the bladder may show an improvement. In patients with pre-existing sphincter hypertonia, acute retention of urine may occur; the drug should be used with caution in such cases.

Laboratory tests

In rare instances, elevated aspartate aminotransferase, blood alkaline phosphatase and blood glucose levels in the serum have been recorded. Appropriate laboratory tests should therefore be performed periodically in patients with liver disease or diabetes mellitus in order to ensure that no drug-induced changes in these underlying diseases have occurred.

Abrupt discontinuation

Anxiety and confusional state, delirium, hallucination, psychotic disorder, mania or paranoia, convulsion (status epilepticus), dyskinesia, tachycardia, hyperthermia, rhabdomyolysis and – as a rebound phenomenon - temporary aggravation of spasticity have been reported following the abrupt withdrawal of Musafen Tablet, especially after long-term medication. Drug withdrawal reactions including postnatal convulsions in neonates have been reported after intrauterine exposure to oral Musafen Tablet. As a precautionary measure, Musafen Tablet administration to neonates with gradual tapering can help in controlling and preventing the withdrawal reactions. (see section USE IN PREGNANCY AND LACTATION).

Withdrawal of baclofen following long-term use (longer than 2 to 3 months) must be gradual, extending over the course of about 3 weeks, except in overdose-related emergencies or where serious adverse effects have occurred, the treatment should always be gradually discontinued by successively reducing the dosage (over a period of approximately 1 to 2 weeks).

Driving and using machines

Musafen Tablet may be associated with adverse effects such as dizziness, sedation, somnolence and visual impairment which may negatively affect the patient's reaction times. Patients experiencing these adverse reactions should be advised to refrain from driving or using machines.

Posture and balance

Musafen Tablet should be used with caution when spasticity is needed to sustain an upright posture and balance in locomotion (see section RECOMMENDED DOSAGE).

INTERACTION WITH OTHER MEDICAMENTS:

Observed interactions to be considered

Levodopa/Dopa Decarboxylase (DDC) inhibitor (Carbidopa)

In patients with Parkinson's disease receiving treatment with Musafen Tablet and levodopa (alone or in combination with DDC inhibitor, carbidopa), there have been reports of mental confusion, hallucinations, headaches, nausea and agitation. Worsening of the symptoms of Parkinsonism has also been reported. Hence, caution should be exercised during co administration of Musafen Tablet and levodopa/carbidopa.

Drugs causing Central Nervous System (CNS) depression

Increased sedation may occur when Musafen Tablet is taken concomitantly with other drugs causing CNS depression including other muscle relaxants (such as tizanidine), with synthetic opiates or with alcohol (see Driving and using machines under section WARNINGS & PRECAUTIONS). The risk of respiratory depression is also increased. In addition, hypotension has been reported with concomitant use of morphine and intrathecal baclofen. Careful monitoring of respiratory and cardiovascular functions is essential, especially in patients with cardiopulmonary disease and respiratory muscle weakness.

Antidepressants

During concomitant treatment with tricyclic antidepressants, the effect of Musafen Tablet may be potentiated, resulting in pronounced muscular hypotonia.

Lithium

Concomitant use of oral Musafen Tablet and lithium resulted in aggravated hyperkinetic symptoms. Thus, caution should be exercised when Musafen Tablet is used concomitantly with lithium.

Antihypertensives

Since concomitant treatment with antihypertensives is likely to enhance the fall in blood pressure, the dosage of antihypertensive medication should be adjusted accordingly.

Agents reducing renal function

Drugs or medicinal products that can significantly impact renal function may reduce baclofen excretion leading to toxic effects (see section WARNINGS & PRECAUTIONS).

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: There are no adequate and well-controlled studies in pregnant women. Animal data showed that baclofen crosses the placental barrier. Therefore, baclofen should not be used during pregnancy unless the expected benefit outweighs the potential risk to the fetus. Drug withdrawal reactions including postnatal convulsions in neonates have been reported after intra-uterine exposure to oral baclofen.

Use in Lactation: In mothers taking Baclofen tablet at therapeutic doses, the active substance passes into the breast milk, but in quantities so small that no adverse effects are to be expected in the infant.

Fertility: There is no data available on the effect of baclofen on human fertility. Baclofen did not impair male or female fertility in rats at dose levels not toxic to them.

SIDE EFFECTS:**Adverse effects usually occur: -**

Adverse effects occur mainly at the start of treatment (e.g., sedation, somnolence), if the dose is increased too rapidly, or if large doses are used. They are often transitory and can be attenuated or eliminated by reducing the dosage; they are rarely severe enough to require stopping the medication. In patients with a history of psychiatric illness or with cerebrovascular disorders (e.g., stroke), as well as in elderly patients, adverse reactions may be more serious. Lowering of the convulsion threshold and convulsions may occur, particularly in epileptic patients. Certain patients have shown increased muscle spasticity as a paradoxical reaction to the medication. Many of the side effects reported are known to occur in association with the underlying conditions being treated.

Central Nervous System: Particularly at the start of treatment, unwanted effects such as daytime sedation, drowsiness and nausea may occur frequently.

Dryness of the mouth, respiratory depression, lightheadedness, lassitude, exhaustion, mental confusion, dizziness, retching, vomiting, personality changes, vertigo, headache and insomnia may be encountered occasionally.

Neurological and/or psychiatric manifestations which have been reported occasionally or rarely include: euphoria, depressive states, paraesthesiae, myalgia, muscular weakness, ataxia, tremor, tinnitus, nystagmus, accommodation disorders, hallucinations, nightmares, dysarthria. It is often difficult to distinguish whether these are drug effects or manifestations of the disease under treatment. Psychiatric manifestations can occur in acute or chronic toxicity due to baclofen.

Lowering of the convulsion threshold and convulsions may occur, particularly in epileptic patients.

Gastrointestinal tract: Frequent: nausea.

Occasional mild gastrointestinal disturbances (constipation, diarrhoea, retching, vomiting, taste disturbances, colicky abdominal pain, nausea).

Cardiovascular system: Occasional hypotension, diminished cardiovascular functions.

Urogenital System: Occasional or rare dysuria, frequency of micturition, enuresis or retention of urine, impotence.

Miscellaneous Unwanted Effects: In rare or isolated cases visual disturbances, hyperhidrosis, rash or disorders of hepatic function may occur.

Certain patients have shown increased spasticity as a paradoxical reaction to the medication.

If an undesirable degree of muscular hypotonia occurs, making it more difficult for patients to walk or fend for themselves, this can usually be relieved by adjusting the dosage (ie. by reducing the doses given during the day and possibly increasing the evening dose).

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms: Prominent features or signs of central nervous depression: Drowsiness, impairment of consciousness, coma and respiratory depression. The following symptoms may also occur: confusion, hallucinations, agitation, convulsions, accommodation disorders, absent pupillary reflex, generalized muscular hypotonia, myoclonus, hyporeflexia or areflexia, peripheral vasodilatation, hypotension or hypertension, bradycardia, tachycardia or cardiac arrhythmia, hypothermia, nausea, vomiting, diarrhea, salivary hypersecretion, increased hepatic enzymes, sleep apnea, rhabdomyolysis, tinnitus and abnormal electroencephalogram (burst suppression pattern and triphasic waves).

A deterioration of the overdose syndrome may occur if various substances or drugs acting on the central nervous system (eg. alcohol, diazepam, tricyclic antidepressants), have been taken at the same time.

Treatment: No specific antidote is known.

Supportive measures and symptomatic treatment should be given for complications such as hypotension, hypertension, convulsions, gastrointestinal disturbances, and respiratory or cardiovascular depression.

Since the drug is excreted chiefly via the kidneys, generous quantities of fluid should be given, possibly together with a diuretic. Hemodialysis (sometimes unscheduled) may be useful in cases of severe poisoning associated with renal failure (see section WARNINGS & PRECAUTIONS).

STORAGE CONDITIONS:

Store below 30°C. Protect from light.

Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZE:

Pack in blister pack of 5 x 10's tablets/ box.

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.